

Measurement of Isolated-Prompt Photons in $p + p$ Collisions at $\sqrt{s} = 200$ GeV with the sPHENIX Detector

A Faithful, Reader-Friendly Rewrite of the Analysis Note

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Abstract

This note presents the first measurement of the isolated-prompt-photon differential cross section in $p + p$ collisions at a center-of-mass energy $\sqrt{s} = 200$ GeV recorded by the sPHENIX detector at RHIC during Run 2024. Photons are measured within a pseudorapidity acceptance $|\eta^\gamma| < 0.7$ and a transverse-energy range $12 < E_T^\gamma < 32$ GeV. The isolation requirement is defined at truth (generator) level as a total isolation energy $E_{\text{iso}}^{\text{truth}} < 4$ GeV inside a cone of radius $\Delta R = 0.3$ around the photon. The measurement uses an integrated luminosity of $\mathcal{L} = 64.37 \text{ pb}^{-1}$. The cross section is corrected for background, detector efficiency and resolution (via iterative Bayesian unfolding) and compared to next-to-leading-order pQCD predictions from JETPHOX and to PYTHIA-8. This rewrite preserves every section, table and numerical result of the original note while expanding the prose for a reader with a high-energy-physics graduate background.

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1 Introduction

The production of prompt photons in hadronic collisions is a classic probe of perturbative quantum chromodynamics (pQCD) and of the parton structure of the proton. Unlike hadrons, a prompt photon emerges directly from the hard scattering and does not fragment or hadronize; it therefore carries clean kinematic information about the underlying partonic process. “Prompt” photons are conventionally separated into two contributions:

- **Direct photons**, produced in the hard $2 \rightarrow 2$ scattering itself. At RHIC energies the two dominant subprocesses are *quark–gluon Compton scattering* $qg \rightarrow q\gamma$ and *quark–antiquark annihilation* $q\bar{q} \rightarrow g\gamma$. The Compton channel involves a gluon in the initial state and therefore makes the cross section directly sensitive to the gluon parton distribution function (PDF) of the proton.
- **Fragmentation photons**, radiated collinearly from a final-state quark or gluon. These are described by photon fragmentation functions and are suppressed by requiring the photon to be *isolated*.

The original note’s Fig. 1 shows the representative leading-order Feynman diagrams for these processes: the Compton diagram ($qg \rightarrow q\gamma$), the annihilation diagram ($q\bar{q} \rightarrow g\gamma$), and a fragmentation diagram in which a photon is emitted from a scattered parton.

Because the Compton process dominates at RHIC, isolated-photon production probes the gluon density of the proton. The momentum fraction carried by the struck partons is approximately

$$x \approx \frac{2 E_T^\gamma}{\sqrt{s}}, \quad (1)$$

so the kinematic range of this measurement, $12 < E_T^\gamma < 32$ GeV at $\sqrt{s} = 200$ GeV, corresponds to $x \approx 0.1$ – 0.3 . This is precisely the region where the gluon PDF is comparatively poorly constrained, making the measurement valuable as a future input to global PDF fits.

Isolation. An *isolation* criterion is imposed to suppress the fragmentation contribution and the much larger background from neutral-meson decays (primarily $\pi^0 \rightarrow \gamma\gamma$ and $\eta \rightarrow \gamma\gamma$). One sums the energy in a cone of fixed radius around the photon, excluding the photon itself, and requires this isolation energy to be small. The original note’s Figs. 2 and 3 illustrate, from simulation, how the fraction of photons passing the isolation cut evolves with E_T^γ and how it differs between signal direct photons and the meson-decay background.

Prior measurements. The PHENIX experiment previously measured the isolated direct-photon cross section in $p+p$ at $\sqrt{s} = 200$ GeV (Phys. Rev. D **86**, 072008, Ref. [1]). sPHENIX, with its hermetic electromagnetic and hadronic calorimetry, extends the reach to higher E_T^γ with substantially improved jet/photon energy resolution, and the present note constitutes the first such sPHENIX result.

2 Data and Monte Carlo Samples

2.1 Data sample

The analysis uses $p+p$ collision data recorded by sPHENIX during the Run 2024 (Run 24) data-taking period. The relevant run range, software production tag and calibration database (CDB) global tag are summarized in Table 1.

Table 1: Data sample definition.

Quantity	Value
Run range	47289–53880
Analysis build	ana521
Calibration global tag (CDB)	2025p007
DST datasets	DST_FITTING, DST_JETCALO

2.2 Monte Carlo samples

Signal and background processes are generated with PYTHIA-8 (Detroit tune, version 8.307) and passed through a full GEANT-4 simulation of the sPHENIX detector. To obtain statistics across the full E_T^γ spectrum, samples are generated in separate hard-process bins: photon-triggered samples with parton-level thresholds of 5, 10 and 20 GeV, and QCD dijet samples with thresholds of 8, 12, 20, 30 (and 40) GeV, each with $\mathcal{O}(10^7)$ events.

Each generated sample carries an effective cross section that accounts for the generator-level acceptance of its hard-process selection,

$$\sigma_{\text{eff}} = \sigma_{\text{gen}} \frac{N_{\text{acc}}}{N_{\text{gen}}}, \quad (2)$$

where σ_{gen} is the generated process cross section, N_{gen} the number of generated events and N_{acc} the number accepted into the sample. The effective cross sections are listed in Table 2.

Table 2: Effective cross sections σ_{eff} [pb] of the MC samples (MDC2 single-interaction / double-interaction productions).

Sample	σ_{eff} [pb]
jet 8	4.93×10^6
jet 12	1.49×10^6
jet 20	6.26×10^4
jet 30	2.53×10^3
photon 5	1.46×10^5
photon 10	6.94×10^3
photon 20	1.30×10^2

To avoid double-counting when stitching the threshold-binned samples together, each sample is restricted to a truth p_T window (the photon or leading-jet truth p_T), given in Table 3.

2.3 Beam crossing angle and double interactions

The Run 24 data were taken under two distinct beam configurations distinguished by the crossing angle, which strongly affects the longitudinal width of the luminous region and hence the rate of double (pileup) interactions:

- **0 mrad crossing angle:** runs 47289–51274, $\mathcal{L} \approx 47 \text{ pb}^{-1}$, $\sigma_z \approx 54 \text{ cm}$, with a double-interaction (DI) fraction of about 22% (precisely 22.4%).

Table 3: Truth p_T windows used to stitch the threshold-binned MC samples.

Sample	Truth p_T window [GeV]
photon 5	[0, 14)
photon 10	[14, 22)
photon 20	[22, ∞)
jet 8	[9, 14)
jet 12	[14, 21)
jet 20	[21, 32)
jet 30	[32, 42)
jet 40	[42, ∞)

- **1.5 mrad crossing angle:** runs 51274–54000, $\mathcal{L} \approx 17 \text{ pb}^{-1}$, $\sigma_z \approx 22 \text{ cm}$, with a DI fraction of about 8% (precisely 7.9%).

Double interactions are modeled in MC by *blending* single-interaction (SI) and double-interaction templates; this is described in detail in Sec. 5 and Appendix 17.

2.4 Trigger

The analysis relies on the EMCal photon (single-tower / cluster) trigger. A Level-1 trigger update was deployed at run 47289; the nominal trigger is the **Photon 4 GeV** trigger (trigger bit 30). The per-event trigger efficiency is measured with a reference (minimum-bias) trigger using the ratio

$$\varepsilon_{\text{trig}} = \frac{N(\text{scaled}[10] \ \& \ \text{live}[N_{\text{bit}}])}{N(\text{scaled}[10])}, \quad (3)$$

i.e. the fraction of minimum-bias-selected events that also fire the photon trigger bit. The efficiency turn-on as a function of E_T is fit with a Gumbel function,

$$\varepsilon_{\text{trig}}(E_T) = p_0 \exp \left[- \exp \left(- \frac{E_T - \mu}{\beta} \right) \right], \quad (4)$$

reaching a plateau efficiency of 0.9966 in the measured E_T^γ range.

2.5 Event selection and luminosity

Events are required to have a reconstructed minimum-bias-detector (MBD) vertex satisfying $|z_{\text{MBD}}| < 60 \text{ cm}$. The integrated luminosity for the analyzed sample is

$$\mathcal{L} = 64.37^{+5.88}_{-4.34} \text{ pb}^{-1} \quad (+9.13\% / -6.75\%), \quad (5)$$

determined from a van der Meer scan with an MBD reference cross section $\sigma_{\text{MBD}} = 25.2^{+2.3}_{-1.7} \text{ mb}$ (see Sec. 6 and the PPG-10 luminosity note, Ref. [10]).

3 Photon Reconstruction and Identification

3.1 Calorimeters

The photon reconstruction relies on the sPHENIX electromagnetic calorimeter (EMCal) and on the inner and outer hadronic calorimeters (iHCal, oHCal). The EMCal is a SPACAL-type tungsten/scintillating-fiber sampling calorimeter with full 2π azimuthal coverage and pseudorapidity coverage $|\eta| < 1.1$.

It is segmented into towers of size $\Delta\eta \times \Delta\phi \approx 0.024 \times 0.024$ (96×256 towers in total) and provides a single-photon energy resolution of $\sigma_E/E \simeq 16\%/\sqrt{E[\text{GeV}]} \oplus 5\%$ at test-beam energies. As noted in Sec. 5, an additional smearing is applied to the EMCal energies in the MC for the nominal analysis to better match the in-situ resolution. The two HCal sections sit immediately behind the EMCal at increasing radius; they provide hadronic shower containment and an energy reading that enters the hadronic-leakage variable R_{had} and the cluster isolation. The collision vertex is reconstructed from the Minimum Bias Detector (MBD), two arrays of 64 quartz-Cherenkov photomultipliers in the forward regions $3.51 < |\eta| < 4.61$.

3.2 Clustering algorithm

EMCal towers are aggregated into clusters by the `RawClusterBuilderTemplate` algorithm. Edge-adjacent towers above a single 70 MeV threshold that pass the `isGood` tower-quality cuts are grouped together; clusters grow until no adjacent tower exceeds the threshold, after which a cluster-splitting algorithm separates overlapping showers. The cluster kinematics use the tower positions in x - y - z space with shower-depth corrections, and the resulting position is converted to η and ϕ relative to the MBD-reconstructed collision vertex; the cluster energy is the scalar sum of the constituent tower energies. The dependence of the cluster η on the vertex z position is the mechanism by which the double-interaction vertex shift (Appendix J) distorts the reconstructed shower shapes.

3.3 Photon energy response

The energy scale and resolution are estimated from MC. The ratio of reconstructed cluster E_T to truth photon E_T^{truth} is studied as a function of E_T^{truth} and fit with a Gaussian in each truth-energy bin (original note's Figs. 8–10). To isolate the intrinsic detector resolution from vertex-induced distortions, this study restricts the vertex to $|z_{\text{vtx}}| < 30$ cm; the wider analysis fiducial $|z_{\text{vtx}}| < 60$ cm is restored in the unfolding response matrix and the efficiency calculation. The fitted mean is close to unity ($\mu \approx 0.99$) and the fractional resolution σ improves from about 0.071 at 8–10 GeV to about 0.051 at 36–45 GeV. A low-side tail reflects residual position-dependent calibration and is absorbed into the unfolding response matrix rather than corrected separately. The signal-to-background ratio of the reconstructed sample rises from $\sim 15\%$ near 10 GeV to $\sim 50\%$ near 30 GeV, which is the underlying reason the extracted purity (Sec. 4) improves so strongly with E_T^γ .

3.4 Shower-shape variables

A set of EMCal shower-shape observables is used to distinguish single photons from the broader and more irregular showers produced by merged $\pi^0 \rightarrow \gamma\gamma$ decays and by hadronic background. These variables are summarized in Table 4.

3.5 Non-physics-background (NPB) rejection

A first XGBoost boosted-decision-tree (BDT) discriminant removes *non-physics background* (NPB): out-of-time energy depositions, noise and beam-induced backgrounds. The NPB BDT uses 25 input features and is trained using the cluster timing as a label: clusters with $t_{\text{cluster}} - t_{\text{MBD}} < -7$ ns are tagged as NPB. Training and evaluation use $6 < E_T < 40$ GeV and $|\eta| < 0.7$. The nominal working point is $\text{NPB} > 0.5$, which gives roughly 99% purity at roughly 99% signal retention; the cut is varied to 0.3 and 0.7 for systematics.

Table 4: Shower-shape variables used in photon identification.

Variable	Description
$E_{3\times 2}/E_{3\times 5}$	ratio of energy in 3×2 to 3×5 tower windows
$E_{1\times 1}/E$	central-tower energy fraction
$w_{\text{COGX}\eta}$	shower width (center-of-gravity) in η
$w_{\text{COGX}\phi}$	shower width (center-of-gravity) in ϕ
w_{72}	7×2 lateral width
R_{had}	hadronic-leakage ratio (HCal behind cluster)
E_1-E_4	ring energies around the cluster core
$E_{t1}-E_{t4}$	corresponding transverse ring energies
BDT	combined boosted-decision-tree discriminant

3.6 Preselection and photon-ID BDT

After NPB rejection, photons pass a loose *preselection* and are then classified by a dedicated photon-ID BDT. The preselection and the tight/non-tight BDT working-point definitions are given in Table 5. The non-tight selection is used as the control sample in the data-driven purity (ABCD) method (Sec. 4).

Table 5: Preselection requirements and tight / non-tight photon-ID working points. The tight and non-tight thresholds are linear in E_T^γ .

Selection	Requirement
<i>Preselection</i>	
central fraction	$E_{1\times 1}/E_{3\times 3} < 0.98$
ring 1 transverse	$0.6 < e_{t1} < 1.0$
window ratio	$0.8 < E_{3\times 2}/E_{3\times 5} < 1.0$
NPB BDT	$\text{NPB} > 0.5$
<i>Tight</i>	
photon-ID BDT	$\text{BDT} > 0.8156 - 0.00156 E_T^\gamma$
<i>Non-tight</i>	
photon-ID BDT	$0.7333 - 0.01333 E_T^\gamma < \text{BDT} < 0.6844 + 0.00156 E_T^\gamma$

The photon-ID BDT (Table 6) uses 11 input features and consists of approximately 750 trees of depth 5.

3.7 Isolation

Several isolation definitions were evaluated by their ROC area-under-curve (AUC); the best performer is a TopoCluster isolation in a cone of radius $R = 0.4$, with AUC in the range ~ 0.82 – 0.88 . The nominal 80%-efficiency isolation working point is

$$E_{\text{iso}} < 0.490 + 0.0370 E_{\text{reco}} \text{ GeV}, \quad (6)$$

and the complementary non-isolated control region is

$$E_{\text{iso}} > 0.8 + 0.490 + 0.0370 E_{\text{reco}} \text{ GeV}. \quad (7)$$

Table 6: Input features of the photon-ID BDT (11 features).

#	Feature
1	E_T
2	$w_{\text{COGX } \eta}$
3	$w_{\text{COGX } \phi}$
4	z_{vtx}
5	η
6	$E_{1 \times 1} / E_{3 \times 3}$
7–10	$E_{t1} - E_{t4}$
11	$E_{3 \times 2} / E_{3 \times 5}$

The isolation cone is referenced to the EMCal radius of 93.5 cm.

4 Analysis Procedure

The measurement proceeds through an eight-step workflow: (1) event and photon selection; (2) NPB rejection; (3) photon-ID and isolation classification into ABCD regions; (4) data-driven purity extraction with signal-leakage correction; (5) MC closure of the purity method; (6) efficiency determination (reconstruction, ID, isolation, MBD-vertex); (7) iterative Bayesian unfolding; and (8) normalization to the differential cross section.

4.1 Truth isolation

At generator level a photon is defined as isolated if the total isolation energy inside a cone $\Delta R = 0.3$ around the photon satisfies $E_{\text{iso}}^{\text{truth}} < 4$ GeV. This truth definition is the reference to which the unfolded result is corrected, decoupling the published cross section from the detector-level reconstruction (TopoCluster $R = 0.4$) isolation.

4.2 Data-driven purity: the double-sideband (ABCD) method

The signal purity is extracted with a two-dimensional double-sideband (“ABCD”) method using the photon-ID (tight/non-tight) and isolation (isolated/non-isolated) axes. The four regions are:

- **A**: tight & isolated (signal region),
- **B**: tight & non-isolated,
- **C**: non-tight & isolated,
- **D**: non-tight & non-isolated.

Assuming the ID and isolation variables are uncorrelated for background, the background in region A is estimated from the sidebands, giving the signal yield

$$N_A^{\text{sig}} = N_A - N_B \frac{N_C}{N_D}. \quad (8)$$

A residual signal-leakage correction accounts for genuine signal photons leaking into the control regions (estimated from MC fractions f_X^{MC} , with $X \in \{B, C, D\}$):

$$N_A^{\text{sig}} = \frac{N_A - N_B \frac{N_C}{N_D}}{1 - (f_B - f_D \frac{N_C}{N_D}) - \dots}, \quad (9)$$

(schematically; the full leakage-corrected expression uses the MC signal fractions in each region). The purity is then

$$P = \frac{N_A^{\text{sig}}}{N_A}. \quad (10)$$

The extracted purity rises strongly with E_T^γ :

- $P = 0.51 \pm 0.01$ at 10–12 GeV,
- $P = 0.72 \pm 0.04$ at 18–20 GeV,
- $P = 0.90 \pm 0.05$ above 26 GeV.

4.3 MC closure and isolation-template tuning

The ABCD procedure is validated by a closure test in MC, C_{MC} , in which the method is applied to simulation and the recovered purity is compared to the true signal fraction. To make the simulated isolation distribution match data, the MC isolation energy is corrected with a linear “fudge” transformation

$$E_{\text{iso}}^{\text{MC,corr}} = 1.2 E_{\text{iso}}^{\text{MC}} + 0.1 \text{ GeV}. \quad (11)$$

4.4 Efficiencies

Four efficiency factors are evaluated from MC as functions of the truth photon E_T^γ and applied bin-by-bin (together with the purity) to correct the measured yields. Truth photons are matched to reconstructed clusters using the track-ID-based association from the `CaloRawClusterEval` module, which assigns to each cluster the truth primary particle whose GEANT hits contribute the largest summed energy across the cluster’s towers; a truth photon is matched when its track ID equals the cluster’s stored leading-energy truth ID. The four factors are defined as follows.

- **Reconstruction efficiency:** the fraction of truth signal photons matched to a reconstructed cluster.
- **Identification efficiency:** the fraction of truth-matched signal photons that also pass the preselection and the tight photon-ID, relative to all truth-matched signal photons.
- **Isolation efficiency:** the fraction of truth-matched signal photons that also satisfy the isolation requirement, relative to all truth-matched signal photons.
- **MBD-vertex efficiency:** the fraction of truth-isolated signal photon events in which the MBD north–south coincidence fires and a reconstructed vertex is found within $|z_{\text{reco}}| < 60$ cm. The denominator uses the all- z truth fiducial, so this factor converts the beam-delivered luminosity into events with a usable reconstructed vertex.

The reconstruction, ID and isolation efficiencies are each in the 0.7–0.9 range and vary slowly with E_T^γ ; the MBD-vertex efficiency is the factor studied for channel independence in Appendix I. The efficiencies are applied after unfolding, on the truth- E_T^γ grid.

4.5 Unfolding

The detector-level spectrum is corrected back to truth level with the D’Agostini iterative Bayesian method as implemented in RooUnfold 3.0.5. The iterative method is preferred over matrix inversion or SVD for two reasons: it converges rapidly when the response matrix is close to diagonal — as it is here, given the $\sim 5\%$ EMCal energy resolution — and the regularization is set transparently

by the iteration count, with each additional iteration trading residual bias against statistical noise. The fiducial binning of the reported result is

$$[12, 14, 16, 18, 20, 22, 24, 26, 28, 32] \text{ GeV}. \quad (12)$$

The response matrix is built on a wider grid that adds a $[10, 12]$ GeV reco-side underflow and a $[32, 36]$ GeV reco-side overflow, paired with truth bins $[8, 10]$ (low underflow) and $[36, 45]$ GeV (high overflow); these absorb migration in and out of the reported range and are not themselves reported.

To make the MC E_T^γ prior resemble data, the response matrix is reweighted by the ratio of a single-iteration D’Agostini unfold of the data spectrum (from a baseline run with the prior reweight disabled) to the un-reweighted MC truth prior, both normalized to unit area. The ratio is fit with an order-[2/2] Padé approximant from 10 to 35 GeV (with the rational denominator’s zeros constrained below the $[10, 30]$ GeV clamp range), and the factor is applied photon-by-photon on the truth E_T^γ when populating the response matrix in the next pipeline iteration. Because the energy resolution is small, the effect of unfolding on the spectrum is minor compared with the statistical uncertainties: the relative change between successive iterations drops below the per-bin statistical uncertainty after the first iteration, so the nominal result uses $N_{\text{iter}} = 2$, which keeps the iteration-dependent bias subdominant to other systematics. The ratio of post- to pre-unfolding yield is 2–10% across the E_T^γ range.

Two closure tests validate the procedure. In the *full* closure test the response matrix is built from the entire MC sample and the same events are unfolded, exactly reproducing the truth spectrum. In the *half* closure test the matrix is built from half the sample and applied to the statistically independent other half; the unfolded result agrees with truth to within 2%, consistent with the statistical uncertainty. The iteration scan that sets N_{iter} is propagated as the unfolding-iteration systematic (Sec. 5, item 5.7) and the prior reweighting as item 5.6.

5 Systematic Uncertainties

The systematic uncertainties are grouped by source. Maximum per-bin values (quoted as down/up where asymmetric) are collected in Table 7.

5.1 Energy scale (flat). A $\pm 1.1\%$ shift of the EMCal energy scale produces a maximum cross-section variation of $-10.5\%/ +12.2\%$.

5.1 Energy scale (non-linearity). A non-linearity slope $s = 5 \times 10^{-4}$ gives up to $\pm 7.2\%$.

5.2 Energy resolution. The nominal MC smearing $\sigma_{\text{MC}} = (0.185, 0, 0.040)$ is compared to a data-like $\sigma_{\text{data}} = (0.15, 0.05, 0.05)$, with variations of no extra smearing and $(0.13, 0.08, 0.08)$; the maximum effect is $\pm 9.6\%$ (up to $\pm 21.2\%$ in the most extreme bin).

5.3.1 Tight-ID variation. Shifting the tight BDT threshold by $+0.05$ gives $\pm 10.4\%$.

5.3.2 Non-tight variation. Shifting the non-tight threshold by -0.10 gives $\pm 2.1\%$.

5.3.3 Non-isolation definition. Using `noniso04/noniso10` control regions gives $-5.1\%/ +0.9\%$.

5.3.4 Fit functional form. Using an error-function fit form gives $\pm 0.7\%$.

5.3.5 Fit confidence interval. The purity-fit CI gives $-8.1\%/ +8.2\%$.

5.3.6 MC closure. The purity-method closure gives $\pm 12.0\%$ (largest at low E_T^γ).

5.4.1 Isolation pedestal. $\pm 5.0\%$.

5.4.2 herwig cross-check. Re-deriving efficiencies with HERWIG 7 gives $\pm 5.3\%$.

5.5 MBD vertex. $\pm 6\%$.

5.6 Response reweighting. $\pm 0.8\%$.

5.7 Unfolding iteration scan. $\pm 4.7\%$.

5.8 Double-interaction blending. Best-fit blend fractions $f_{\text{best}} = 0.290$ (0 mrad) and 0.000 (1.5 mrad) give $\pm 1.6\%$.

5.9 NPB cut. Varying the NPB working point gives $-1.4\% / +6.1\%$.

5.10 Luminosity. $+9.13\% / -6.75\%$.

Table 7: Maximum per-bin systematic uncertainties [%] (down / up where applicable).

Source	Max down [%]	Max up [%]
Energy scale (flat)	10.49	11.87
Energy scale (non-lin.)	7.26	7.26
Energy resolution	21.21	21.21
Purity – tight		6.72
Purity – non-tight		2.35
Purity – non-iso	3.80	0.99
Purity – fit form		0.33
Purity – fit stat (CI)		4.40
Purity – MC closure		5.77
Efficiency – isolation		4.73
Efficiency – HERWIG		5.83
MBD vertex		6.00
Unfolding – reweight		0.36
Unfolding – iteration		5.50
Double interaction		1.57
NPB cut	1.03	4.54
Luminosity	6.75	9.13
Total	30.80	31.72

6 Results

The reconstructed yield is obtained by unfolding the background-subtracted, purity-weighted signal,

$$Y_{\text{rec}} = \text{Unfold}(N_{\text{tight,iso}} \cdot P), \quad (13)$$

and the isolated-prompt-photon differential cross section is

$$\frac{d\sigma}{dE_{\text{T}}^{\gamma} d\eta} = \frac{1}{\mathcal{L}} \frac{Y_{\text{rec}}}{\varepsilon \Delta E_{\text{T}}^{\gamma} \Delta \eta^{\gamma}}, \quad (14)$$

where ε is the combined efficiency, ΔE_T^γ the bin width and $\Delta\eta^\gamma = 1.4$ the pseudorapidity acceptance ($|\eta^\gamma| < 0.7$).

The fully corrected differential cross section (original Fig. 54) falls by roughly two orders of magnitude across the reported $12 < E_T^\gamma < 32$ GeV range. It is compared to two theory calculations. PYTHIA-8 (version 8.307, Detroit tune) is run with the same truth-level isolation as the data. JETPHOX (v1.3.1.4) computes the direct+fragmentation cross section at NLO with CT14nlo PDFs, BFG II fragmentation functions, and scales $\mu_R = \mu_F = \mu_f = E_T^\gamma$; an additional NLO prediction without isolation is overlaid for reference. Both calculations agree with the measurement within experimental uncertainties across the full E_T^γ range, supporting the gluon-Compton-dominance picture at the parton momentum fractions probed here.

The measurement is systematics-dominated over the whole range. The statistical uncertainty grows from sub-percent at the lowest reported bin to about 30% at the highest but remains sub-dominant. The energy-scale and double-interaction systematics dominate the high- E_T^γ budget, while the purity (fit-CI and MC-closure) and luminosity sources drive the low- E_T^γ budget; the total systematic ranges from about $\pm 14\%$ at low E_T^γ to $-30.8\% / +31.7\%$ near the top of the range (Table 7).

The result is also compared to the earlier PHENIX measurement (Ref. [1], original Fig. 55), which reports inclusive direct photons (no isolation) in $|\eta| < 0.25$. The PHENIX spectrum is shown both as published and after the bin-width recovery and pseudorapidity-density rescaling of Appendix H; the corrected spectrum is consistent with the present measurement within uncertainties across the overlap range, with the small residual offset attributable to the fragmentation contribution that survives the truth-level isolation cut.

7 Conclusion

This note presents the first sPHENIX measurement of the isolated-prompt-photon differential cross section in $p + p$ collisions at $\sqrt{s} = 200$ GeV, for $|\eta^\gamma| < 0.7$ and $12 < E_T^\gamma < 32$ GeV, with a truth isolation $E_{\text{iso}}^{\text{truth}} < 4$ GeV in $\Delta R = 0.3$ and an integrated luminosity of 64.37 pb^{-1} . The result is corrected for background via a data-driven double-sideband purity method, for efficiency, and for detector resolution via iterative Bayesian unfolding, and is found to be consistent with NLO pQCD (JETPHOX) and with PYTHIA-8, as well as with the earlier PHENIX measurement. The measurement demonstrates the photon reconstruction and identification capabilities of the sPHENIX calorimeter system and provides a foundation for future gluon-PDF-sensitive measurements at RHIC.

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- [9] T. Adye, RooUnfold.
- [10] sPHENIX PPG-10 luminosity note.
- [11] ATLAS electron/photon performance, JINST **14**, P12006.
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Appendices

A NPB purity and efficiency scans

The non-physics-background (NPB) discriminant introduced in Sec. 4 is a dedicated XGBoost classifier whose purpose is to remove energy depositions in the EMCal that are not associated with a genuine $p + p$ collision: out-of-time clusters, electronic noise, and beam-induced “streak” clusters that align in a narrow band of constant η or ϕ . These clusters are pathological in that they carry no recoil activity and have shower shapes that lie outside the physical phase space populated by single photons or by hadronic showers. Because they survive into the photon-candidate sample if left untreated, they would inflate the measured yield and bias the purity extraction; the NPB BDT is therefore applied *before* the photon-ID BDT and the isolation classification.

The BDT is trained using cluster timing as the supervisory label. The cluster time is referenced to the MBD event time, and clusters with $t_{\text{cluster}} - t_{\text{MBD}} < -7$ ns are tagged as NPB, since genuine collision clusters are tightly peaked in time around the beam crossing whereas out-of-time and beam-induced deposits populate the early tail. Twenty-five input features are used, and training and evaluation are restricted to the fiducial region $6 < E_{\text{T}} < 40$ GeV and $|\eta| < 0.7$, slightly wider in E_{T} than the analysis window so that the turn-on and the high- E_{T} tail are both well sampled.

The working-point study scans the NPB BDT threshold and records, as a function of E_{T}^{γ} , both the photon purity (the fraction of selected clusters that are genuine physics clusters, estimated from the timing-labeled control sample) and the signal-retention efficiency (the fraction of genuine clusters that survive the cut). The nominal cut $\text{NPB} > 0.5$ achieves approximately 99% purity while retaining approximately 99% of genuine clusters, an unusually favorable operating point that reflects the cleanly separated, almost non-overlapping NPB and physics populations in the timing-trained feature space. To bracket the residual modeling dependence of this choice the cut is loosened to 0.3 and tightened to 0.7; the resulting per-bin variation of the unfolded cross section is propagated as the NPB systematic (Sec. 5, item 5.9), where it is found to be small except at the lowest E_{T}^{γ} bin. An independent, physics-based cross-check of the same BDT using a back-to-back recoil-jet requirement is presented in Appendix G.

B Shower-shape distributions and correlations

This appendix collects the detector-level shower-shape distributions that motivate the variable choices of Tables 4 and 6 and that underpin both the NPB rejection and the photon-ID BDT. The original note’s Figs. 57, 13, 58, 19, 59 and 60 show these distributions for data, signal MC, inclusive (background-dominated) MC, and an NPB-enriched data template, together with their pairwise correlations.

Distributions without preselection. At low E_{T}^{γ} (14–18 GeV, original Fig. 57) the shower-shape variables are shown before any preselection for four overlaid samples: data (black), signal MC (red), inclusive MC (blue), and an NPB-tagged data template (green). The NPB-tagged template is the subset of data selected as streak-like by a high center-of-gravity width $w_{\text{COGX}\eta}$ tag; it is normalized to the bulk data by tail-matching in the region $w_{\text{COGX}\eta} > 1$, where the streak population dominates, and serves as a background-enhanced template for the non-physics component. The comparison demonstrates that the streak clusters occupy the broad high- w tail that is essentially absent in the signal MC, which is precisely what the NPB BDT exploits.

Effect of preselection. The same distributions are shown after the loose preselection (original Fig. 58, here at 10–14 GeV). The preselection suppresses the high-tail population in the variables most sensitive to non-physics backgrounds, $w_{\text{COGX}\eta}$ in particular, while leaving the bulk of the physics-cluster shapes essentially unchanged. The per-panel χ^2/ndf between inclusive MC and data remains large for several variables, a reminder that the inclusive MC is a mixture whose composition (signal fraction, NPB fraction, pileup fraction) is not identical to data; the purity method (Sec. 4) is designed to be robust against exactly this kind of template imperfection, since it extracts the signal yield from the data sidebands rather than from an absolute MC normalization.

Correlation with isolation. It is important that the shower-shape discriminant and the isolation variable be as uncorrelated as possible for background, because the double-sideband (ABCD) purity method assumes factorization of the two axes. The original Figs. 59 and 60 show the mean isolation energy $\langle E_{\text{iso}} \rangle$ as a function of each shower-shape value, for background MC, at low (10–14 GeV) and high (28–30 GeV) E_T^γ . The measured linear correlation coefficients are modest, ranging from about -0.26 to $+0.42$ depending on the variable and the E_T^γ range; the width variables $w_{\text{COGX}\eta}$ and $w_{\text{COGX}\phi}$ are the most strongly (positively) correlated with isolation, while the ring transverse energies $E_{t2}-E_{t4}$ are nearly uncorrelated. These residual correlations are the physical origin of the signal-leakage correction applied to the ABCD result and are quantified by the MC closure test described in Sec. 4.

C Isolation distributions

The reconstructed isolation-energy distribution E_{iso} is the second axis of the double-sideband purity method, and its behavior as a function of E_T^γ and of photon-ID category is what makes the ABCD construction work. The original note’s Fig. 61 shows E_{iso} in six E_T^γ bins spanning 10–30 GeV, overlaying three samples: tight-ID photons in inclusive MC (black), non-tight-ID photons in inclusive MC (red), and tight-ID photons in signal MC (blue).

Several features support the analysis design. First, the tight signal-MC distribution is sharply peaked near $E_{\text{iso}} \approx 0$ with a narrow tail, confirming that genuine isolated prompt photons deposit very little additional energy in the cone; this is what justifies the 80%-efficiency isolation working point $E_{\text{iso}} < 0.490 + 0.0370 E_{\text{reco}}$ GeV. Second, the non-tight inclusive sample is shifted to substantially larger E_{iso} , reflecting the jet-like environment of the meson-decay and fragmentation backgrounds that populate the non-tight ID region; this is the basis for using the non-tight sample as a background-shape control. Third, the separation between the signal peak and the background tail grows with E_T^γ , which is the microscopic reason the extracted purity rises from ~ 0.5 at 10–12 GeV to ~ 0.9 above 26 GeV.

The simulated E_{iso} distribution does not match data perfectly, because the underlying-event activity and the soft-particle spectrum entering the cone are imperfectly modeled. This residual mismatch is corrected with the linear “fudge” transformation $E_{\text{iso}}^{\text{MC,corr}} = 1.2 E_{\text{iso}}^{\text{MC}} + 0.1$ GeV introduced in Sec. 4; the slope and offset are tuned so that the fudge-corrected MC reproduces the data isolation shape in a background-dominated control region. The adequacy of this correction is one of the inputs to the purity MC-closure systematic (Sec. 5, item 5.3.6), and the sensitivity to the precise pedestal value is captured by the isolation-pedestal systematic (item 5.4.1).

D Shower-shape cut comparison

This appendix documents the effect of the loose preselection chain on the shower-shape distributions and, by extension, justifies the choice of a BDT photon identifier over a traditional sequential (rectangular) cut selection. The original note’s Fig. 62 ($10 < E_T^\gamma < 12$ GeV) and Fig. 63 ($26 < E_T^\gamma < 35$ GeV) show, for the inclusive-MC background sample, each shower-shape variable before any cut (blue) and after the full preselection chain of Table 5 (red): the central-fraction requirement $E_{1\times 1}/E_{3\times 3} < 0.98$, the first-ring transverse energy window $0.6 < e_{t1} < 1.0$, the window ratio $0.8 < E_{3\times 2}/E_{3\times 5} < 1.0$, and the NPB score requirement $\text{NPB} > 0.5$.

The qualitative message is that the preselection suppresses the high tails of the variables that are sensitive to non-physics background, most visibly the center-of-gravity width $w_{\text{COGX}\eta}$, while having little effect on the core of the physics-cluster shapes. In other words, the preselection removes the pathological clusters without sculpting the discriminating variables of the genuine population, which is the desired behavior before the multivariate photon-ID step.

The broader motivation for using a BDT rather than a sequential cut is that the shower-shape variables are correlated, so a product of one-dimensional rectangular windows discards discriminating information contained in their joint distribution. The photon-ID BDT (Table 6) combines the eleven inputs into a single discriminant and, evaluated at fixed signal efficiency, yields a higher background rejection than the equivalent rectangular selection. The BDT also allows the working point to depend smoothly on E_T^γ (the tight/non-tight thresholds of Table 5 are linear in E_T^γ), which accommodates the E_T^γ -dependence of both the signal shower shapes and the background composition in a way that fixed cuts cannot.

E ϕ -symmetry tower acceptance

This appendix studies the residual data–MC mismatch in the EMCal dead-tower acceptance and derives a purely data-driven systematic on the cross section from it. The sPHENIX EMCal contains dead and malfunctioning towers whose response is not perfectly emulated by the GEANT-4 simulation. If the set of dead towers active in Run 24 data differs from the set modeled in MC, the MC-derived reconstruction efficiency is biased high in the affected regions, which in turn biases the measured cross section low. The ϕ -symmetry method exploits the fact that the inclusive cluster yield is ϕ -symmetric at fixed η in $p + p$ collisions to derive a dead-tower mask directly from data, without relying on any particular MC dead-tower list. Because the same mask is then applied identically to numerator and denominator on both the data and MC sides, the efficiency-corrected cross section becomes sensitive only to the residual data-versus-MC asymmetry in the masked region, not to the absolute mask coverage.

Method. The EMCal is binned in tower index as $96 (i_\eta) \times 256 (i_\phi)$. The towers are grouped into 48 two-row η bands. For each band the 256 per- ϕ cluster counts are projected out. Under the assumption of ϕ -symmetry, deviations from a flat distribution trace either statistical fluctuations or local detector effects. An iterative 3σ -clipped Gaussian is fit to the 256-bin distribution, yielding a band-by-band mean $\mu_{\phi,\text{band}}$ and width $\sigma_{\phi,\text{band}}$. A tower (i_η, i_ϕ) is flagged as data-deficient when $(n_{\text{data}} - \mu_{\phi,b})/\sigma_{\phi,b} < -2$; both η rows in the band are masked whenever a ϕ bin is flagged. The original note’s Fig. 64 shows an example for the band $i_\eta = 64\text{--}67$ ($\eta \approx 0.37\text{--}0.46$), in which a contiguous block of ϕ cells around $i_\phi \approx 30\text{--}45$ falls well below the $\mu - 2\sigma$ threshold and is tagged.

Four masks are built at successively tighter selection levels: (i) *preselect* ($E_T^\gamma \geq 8$ GeV with no saturation and no shape cuts); (ii) *common* (preselect intersected with the NPB cut, $E_{1\times 1}/E_{3\times 3} <$

0.98, and the width windows); (iii) *tight* (common intersected with the ten shower-shape windows and the parametric BDT requirement); and (iv) *OR* (the union of the common and tight masks). The corresponding masked-tower counts are 1344, 1160, 646 and 1180. The monotonic decrease with selection tightness is the expected statistical behavior: the tighter samples have lower per- ϕ statistics, so the fit width σ is broader and fewer bins fall below the -2σ threshold. The tight mask is largely contained in the common mask (a 626-tower overlap), so the OR mask adds only about 20 towers relative to common alone.

Results. The mask is applied as a cluster-level veto, identically to data and MC, and the full efficiency, merge, and yield pipeline is re-run for each variant. The resulting integrated cross-section shifts, evaluated in $E_T^\gamma \in [10, 36]$ GeV relative to the mask-free nominal $\sigma_{\text{nom}} = 1963$ pb, are collected in Table 8. All four shifts are positive, as expected: masking removes MC content in data-dead regions that was inflating the MC-derived efficiency, so the corrected cross section rises once the asymmetry is removed. The four variants agree to within about 1%, consistent with a uniform detector-level dead-tower effect, and the per- E_T^γ -bin ratios lie within $\pm 3\%$ of unity across $E_T^\gamma \in [10, 30]$ GeV (original Fig. 67). The larger scatter in the highest $[32, 36)$ GeV bin is statistical and does not affect the integrated result, which is dominated by the low- E_T^γ region.

An independent estimator rebins the data and lumi-weighted MC tower maps into 4×4 super-cells and flags super-cells whose $\log(R)$ of the data/MC ratio deviates below a Gaussian fit to the bulk; this tags 2–3% of super-cells per selection level and yields an integrated shift agreeing with the ϕ -symmetry mask to about 1%. The convergence of two independent data-driven estimators on the same $\sim 2\%$ level provides a robust bound that does not rely on any specific MC dead-tower list. The per-bin envelope adopted for the systematic, taken from the OR mask, is $\pm 1.57\%$.

Table 8: ϕ -symmetry tower-acceptance cross-check. Nominal cross section $\sigma_{\text{nom}} = 1963$ pb; shifts for each mask definition.

Mask	Shift [%]
Preselection	+2.01
Common	+2.26
Tight	+1.37
Logical OR	+2.28
Per-bin envelope	± 1.57

F jetphox PDF dependence

The NLO theory reference used for comparison with the measured cross section is JETPHOX (v1.3.1.4) with the CT14nlo parton distribution functions and the BFG II photon fragmentation functions. JETPHOX computes the isolated-photon cross section at next-to-leading order in α_s , including both the direct contribution (the photon produced in the hard matrix element) and the fragmentation contribution (the photon produced in the collinear fragmentation of a final-state parton), with the fragmentation piece evolved through the BFG fragmentation functions. To quantify the PDF-driven spread of the theory prediction — and in particular its sensitivity to the proton gluon density, which the Compton subprocess $qg \rightarrow q\gamma$ makes the dominant initial state at RHIC — the production is rerun with two alternative PDF sets, the LO CT14lo set and the NLO NNPDF3.1 set ($\alpha_s(M_Z) = 0.118$), using identical kinematic and fragmentation settings.

Method. All three productions share the same configuration: $\sqrt{s} = 200$ GeV $p + p$, photon rapidity $|y| < 0.7$, direct+fragmentation components with BFG II, a fixed-cone truth isolation with $R = 0.3$ and $E_{\text{iso}}^{\text{truth}} < 4$ GeV matching the analysis definition, and renormalization/factorization/fragmentation scales $\mu_R = \mu_F = \mu_f = c_m \cdot p_T$ with $c_m \in \{0.5, 1.0, 2.0\}$ to map the scale uncertainty. Per scale point, 100 segments of 10^6 events each are generated, totaling 10^8 events. The full PDF error set is computed for every point (56 CT14 Hessian eigenvector pairs for CT14lo/CT14nlo, 100 Monte Carlo replicas for NNPDF3.1). The three productions are statistically independent: only the LHAPDF identifier differs and the random-number seeds are offset so that no event streams are shared. Pairing the LO CT14lo set with the NLO matrix element is a deliberate LO-in-NLO inconsistency — the missing NLO gluon evolution is absorbed into an effective LO fit that is not strictly valid at all x when convolved with the NLO hard cross section — and is shown here only as a methodological probe of the LO-versus-NLO PDF effect, not as a physical prediction.

Results. Table 9 lists the central-scale ($\mu = 1.0 \cdot p_T$) per-bin cross section for the three PDF sets, with the ratios to CT14lo. Moving from the LO CT14lo set to an NLO set shifts the prediction monotonically in E_T^γ : at low E_T^γ (8–14 GeV) the NLO PDFs lie 3–11% *above* CT14lo; at mid E_T^γ (14–26 GeV) they lie 4–25% *below*; and at the highest bins (26–36 GeV) they lie 27–50% *below*. The crossover near $E_T^\gamma \approx 14$ GeV reflects the well-known LO-versus-NLO gluon difference: at the moderate $x \sim 2E_T^\gamma/\sqrt{s} \sim 0.1$ accessed at low E_T^γ the LO gluon undershoots because NLO K -factors are absorbed into the LO fit, whereas at higher $x \gtrsim 0.2$ the LO fit over-compensates to match jet data with the LO matrix element, an excess that the NLO matrix element times NLO PDF correctly suppresses. The two NLO PDFs (CT14nlo and NNPDF3.1) agree with each other to $\pm 3\%$ at low E_T^γ and grow apart to $\pm 20\%$ in the highest bin, where the PDF-family spread becomes comparable to the renormalization-scale envelope ($\pm 40\%$ from $\mu = 0.5/1.0/2.0$). This confirms that the LO-versus-NLO PDF effect, rather than the specific NLO fit family, is the dominant PDF-driven theory uncertainty.

Table 9: JETPHOX cross section [pb] per E_T^γ bin for three PDF sets and ratios to CT14lo.

E_T^γ [GeV]	CT14lo	CT14nlo	NNPDF3.1	nlo/lo	nnpdf/lo
8–10	1519	1673	1690	1.101	1.113
10–12	539.9	571.2	575.4	1.058	1.066
12–14	218.5	222.2	225.1	1.017	1.030
14–16	98.4	96.2	94.4	0.978	0.959
16–18	47.0	44.0	41.9	0.937	0.891
18–20	23.5	20.8	19.1	0.887	0.815
20–22	9.82	8.60	8.01	0.876	0.816
22–24	6.54	5.53	4.95	0.846	0.756
24–26	3.66	2.93	2.73	0.799	0.745
26–28	2.10	1.41	1.37	0.669	0.651
28–32	0.940	0.608	0.579	0.647	0.616
32–36	0.323	0.236	0.165	0.731	0.510
Integrated 8–36	4942	5294	5330	+7.1%	+7.9%

G Back-to-back jet NPB cross-check

This appendix provides an independent, physics-based validation of the non-physics-background BDT that does not rely on the timing label used to train it. The NPB BDT is trained against beam-induced streak clusters that are not associated with a real $p + p$ collision and therefore carry no recoil jet. By contrast, a genuine direct or fragmentation photon (and a high- E_T π^0 candidate) is produced in a $2 \rightarrow 2$ hard scattering and is accompanied by a balancing parton in the opposite hemisphere. Requiring a back-to-back recoil jet (“b2b jet”) therefore selects a sample that is physically depleted of NPB clusters, independently of any shower-shape-based veto. Comparing the unfolded cross section in this b2b-enriched sample with and without the NPB BDT cut is thus a clean closure test: if the BDT is unbiased on real photons, the two should agree to within statistics.

Method. Four variants are produced on top of the analysis pipeline. `b2bjet_pt5` adds a recoil-jet requirement with $p_T^{\text{jet}} \geq 5$ GeV back-to-back in ϕ ($|\Delta\phi| > 3\pi/4$), keeping the nominal NPB cut $\text{NPB} > 0.5$. `b2bjet_pt5_npb0` is the same but with the NPB cut switched off. `b2bjet_pt7` and `b2bjet_pt7_npb0` repeat these with the higher threshold $p_T^{\text{jet}} \geq 7$ GeV. Each variant runs the full data+MC pipeline (purity, unfolding, efficiency, luminosity normalization) at the same $\mathcal{L} = 64.37 \text{ pb}^{-1}$ as the nominal, and the resulting differential cross section is compared bin-by-bin to the nominal selection.

Results. Two conclusions follow (original Figs. 69 and 70). First, the NPB-on and NPB-off curves of the same recoil-jet threshold overlay tightly across the analysis range: bin-by-bin, $|\sigma_{\text{pt5}}/\sigma_{\text{nom}} - \sigma_{\text{pt5, npb0}}/\sigma_{\text{nom}}| \leq 1.0\%$ and the corresponding $p_T^{\text{jet}} \geq 7$ GeV difference is $\leq 0.5\%$, with the largest values in the lowest bin. The NPB cut therefore does not change the answer once a genuine recoil jet is required, exactly the behavior expected if the BDT is unbiased on real photons. Second, the b2b-enriched cross section sits 6–10% below the nominal in the lowest bin (12–14 GeV) and recovers to within $\pm 10\%$ of nominal for $E_T^\gamma \geq 14$ GeV. The deficit is larger for the $p_T^{\text{jet}} \geq 7$ GeV variant (10.0%) than for the ≥ 5 GeV variant (6.9%), which is the ordering expected from recoil-jet *acceptance*: at low E_T^γ the recoil parton is soft and hadronizes into a low- p_T jet that the b2b cut sometimes rejects, more often at the higher threshold. This interpretation is confirmed directly by the truth-level recoil-jet probability (original Fig. 70), which rises from about 36% (24%) at $E_T^\gamma = 10$ –12 GeV for $p_T^{\text{jet}} \geq 5$ (7) GeV to about 75% (71%) at $E_T^\gamma = 32$ –36 GeV. The low- E_T^γ deficit is therefore a recoil-jet-acceptance effect inherited from the PYTHIA-8 fragmentation modeling, not an NPB-cut bias, and is not propagated as a systematic on the nominal cross section.

H PHENIX comparison corrections

The PHENIX measurement of inclusive direct photons (Ref. [1]) is reported in the rapidity acceptance $|\eta| < 0.25$ as a representative cross section per E_T^γ bin derived from the invariant cross section $E d^3\sigma/dp^3$. To overlay it on the present isolated-photon measurement at $|\eta^\gamma| < 0.7$ on an equal footing (original Fig. 55), two corrections are applied to the published points: a bin-width recovery from a smooth fit, and a pseudorapidity-density rescaling derived from prompt-photon truth simulation. Both corrections preserve the published central values and their fractional statistical uncertainties; they only remove geometric and binning differences between the two experiments.

Bin-width recovery. The PHENIX invariant cross section is first converted to the differential form $d^2\sigma/(d\eta dE_T^\gamma)$ through the Jacobian $2\pi E_T^\gamma$, and the result is fit over the published 5–26 GeV

range with a smooth modified power law

$$f(E_T^\gamma) = A \left(1 + \frac{(E_T^\gamma)^2}{b} \right)^c, \quad A = 4.53 \times 10^5, \quad b = 6.34, \quad c = -3.602, \quad (15)$$

with $\chi^2/\text{ndf} = 28.07/15$ and per-bin pulls bounded by $|2.5|$. Because PHENIX quotes each point at the bin center after its own bin-shift correction, converting to a strict bin average requires only a smooth multiplicative shape factor read off the fit,

$$y_i^{\text{bin-avg}} = y_i^{\text{pub}} \frac{\langle f \rangle_i}{f((E_T^\gamma)_i^c)}, \quad \langle f \rangle_{\text{bin}} = \frac{1}{\Delta E_T^\gamma} \int_{\text{bin}} f(E_T^\gamma) dE_T^\gamma, \quad (16)$$

where $(E_T^\gamma)_i^c$ is the bin center. Only the few-percent shape distortion of a steeply falling spectrum within finite-width bins is removed; the published value and its statistical scatter are preserved.

Pseudorapidity-density rescaling. The two experiments cover different rapidity intervals, so the spectra are placed on a common acceptance via the density ratio

$$R(E_T^\gamma) = \frac{(dN/d\eta)|_{|\eta| < 0.25}}{(dN/d\eta)|_{|\eta| < 0.7}}, \quad (17)$$

evaluated from prompt-photon truth simulation (PYTHIA-8, direct+fragmentation, no isolation requirement) on the analysis truth- p_T grid (original Fig. 72). The ratio is consistent with unity to within 0.5% below $E_T^\gamma \approx 14$ GeV and rises monotonically to $R \approx 1.22$ at $E_T^\gamma = 36\text{--}45$ GeV, reflecting the narrowing of the rapidity distribution of $2 \rightarrow 2$ prompt-photon production at the higher Bjorken- x accessed at high E_T^γ . Each bin-averaged PHENIX value is divided by $R((E_T^\gamma)_i^c)$ to produce the $|\eta| < 0.7$ -equivalent spectrum. Imposing the truth isolation $E_{\text{iso}}^{\text{truth}} < 4$ GeV on the prompt-photon sample shifts $R(E_T^\gamma)$ by less than 0.1% in every bin, so the same factor applies to PHENIX's inclusive fiducial within the precision of either measurement.

I MBD photon/jet cross-check

The MBD-vertex efficiency — the probability that the minimum-bias detector reconstructs a valid event vertex, $z_{\text{reco}} \neq -9999$ — enters the cross section as one of the four efficiency factors of Sec. 4. Because this efficiency depends on the forward particle activity that fires the MBD, it could in principle differ between the photon-triggered topology and a generic jet topology. This appendix verifies that it does not, which justifies transferring the MBD systematic between channels (Sec. 5, item 5.5).

The MBD-vertex-found fraction is computed in PYTHIA-8 as a function of the leading truth p_T , separately for the photon channel (the `photon 5/10/20` samples, using the leading truth-photon p_T) and for the jet channel (the `jet 5/8/12/20/30/40` samples, using the leading $R = 0.4$ truth-jet p_T). Per-sample truth- p_T windows (Table 3) prevent double-counting across sample boundaries, and each event carries the weight $\sigma_i/N_i^{\text{gen}}$. The two channels overlay to within 1% across the full reported range (original Fig. 73) and fall monotonically from about 0.80 at $p_T^{\text{lead}} = 5$ GeV to about 0.66 at 35 GeV. The decline reflects the reduced forward activity at higher Bjorken- x : as the hard scale rises, more of the collision energy is carried into the central region and less into the forward MBD acceptance, lowering the coincidence probability. The channel-level agreement demonstrates that the MBD-vertex efficiency is governed by the underlying-event and beam-remnant activity rather than by the hard-process flavor, which is what permits the data-driven MBD efficiency (derived in the PPG-10 luminosity note) to be applied to the photon measurement.

J Double-interaction (pileup) modeling

At the luminosities reached in Run 24 a non-negligible fraction of bunch crossings contain more than one $p + p$ collision. A second, overlapping collision affects the photon measurement in two distinct ways: it adds soft energy that can enter the isolation cone, and — more importantly — it biases the reconstructed event vertex, which in turn shifts the inferred cluster pseudorapidity and distorts the shower-shape inputs to the photon-ID BDT. This appendix develops the full double-interaction (DI) model used throughout the analysis. Rather than applying a factorized after-the-fact correction, the analysis builds dedicated full-GEANT DI samples and *blends* them with the single-interaction (SI) samples at the physically determined cluster fraction, so that the efficiency, the response matrix, and the background template all inherit the pileup effects consistently.

DI samples. Dedicated `photon{5,10,20}_double` and `jet{8,12,20,30,40}_double` samples are produced by full simulation of two overlapping $p + p$ collisions: one hard-scatter event drawn from the same PYTHIA-8 generator as the corresponding SI sample, plus one minimum-bias PYTHIA-8 (Detroit-tune) event overlaid as pileup. The two collisions are placed at independently drawn primary vertices from the generator beam profile. These DI samples are mixed with the nominal SI samples at the cluster-weighted fractions of Table 10, providing both the DI efficiency study and the shower-shape data–MC validation below.

Pileup fractions. The mean number of interactions per crossing is obtained from the Poisson-corrected MBD coincidence rate,

$$\mu = \frac{\sum_{\text{runs}} N_{\text{MBD,corr}}}{r_B \sum_{\text{runs}} \Delta t}, \quad r_B = 111 \times 78\,200 = 8.68 \times 10^6 \text{ Hz}, \quad (18)$$

where r_B is the RHIC bunch-crossing rate. The event-level interaction fractions follow a zero-truncated Poisson (one collision is guaranteed by the trigger),

$$f_1 = \frac{\mu e^{-\mu}}{1 - e^{-\mu}}, \quad f_2 = \frac{(\mu^2/2)e^{-\mu}}{1 - e^{-\mu}}, \quad f_{\geq 3} = 1 - f_1 - f_2, \quad (19)$$

computed per run and luminosity-weighted. Because multi-interaction events carry a higher cluster multiplicity, the relevant quantities for the cluster-level blend are the cluster-weighted fractions

$$w_{\text{single}} = \frac{f_1}{\sum_k k f_k}, \quad w_{\text{double}} = \frac{2f_2 + 3f_{\geq 3}}{\sum_k k f_k}, \quad (20)$$

with triple-and-higher interactions folded into the double bin (they are rare enough that the two-collision sample models them adequately). The resulting fractions and weights are given in Table 10; the cluster-weighted w_{double} values (0.224 at 0 mrad, 0.079 at 1.5 mrad) are the numbers used throughout the analysis.

Vertex-shift mechanism. In a DI event the MBD reconstructs a single vertex whose best estimator is the average of the two collision points,

$$z_{\text{reco}} \approx \frac{z_1 + z_2}{2}. \quad (21)$$

The two true vertices are drawn from the luminous region (data) or the generator profile (MC), so the displacement between the primary truth vertex and the reconstructed vertex, $\Delta z \equiv z_{\text{reco}} - z_{\text{truth}}$,

Table 10: Double-interaction parameters by crossing-angle configuration.

Quantity	0 mrad	1.5 mrad
μ_{corr}	0.24	0.08
f_1	0.879	0.960
f_2	0.111	0.039
w_{single}	0.776	0.921
w_{double}	0.224	0.079

can reach several tens of cm. A cluster reconstructed at the EMCal barrel radius $R_{\text{EMCal}} = 93.5$ cm then acquires a pseudorapidity shift that follows directly from the geometry,

$$\Delta\eta \approx -\frac{\Delta z}{R_{\text{EMCal}} \cosh \eta}, \quad (22)$$

reducing at $\eta = 0$ to $|\Delta\eta| \approx |\Delta z|/93.5$ cm. This η shift is the root cause of the shower-shape distortion that drives the photon-ID degradation below. The RMS of Δz grows from 7.6 cm in SI MC to 34.5 cm in DI MC (Table 11).

A per-event check confirms that the second collision does not add genuine prompt-photon content: the mean number of truth photons per event is 0.493 (single) versus 0.495 (double), and the mean truth isolation energy is 0.17 versus 0.26 GeV, both far below the 4 GeV fiducial threshold. The DI effect is therefore a reconstruction effect, not a change in the underlying signal.

Per-stage efficiency. Truth-to-reco association uses the GEANT track-ID from the `CaloRawClusterEval` module as the unambiguous ground truth, with no additional geometric requirement (so that a fully reconstructed but η -shifted cluster is still counted as found). The factorized chain $\varepsilon_{\text{reco}} \rightarrow \varepsilon_{\text{iso}} \rightarrow \varepsilon_{\text{id}} \rightarrow \varepsilon_{\text{all}}$ then isolates the stage at which each pileup effect acts; the integrated values for SI, DI, and the 0 mrad mixed sample are given in Table 12. The reconstruction stage loses 20 percentage points (pp), of which about 5.5 pp is genuine cluster loss (no track-ID match: 7.9% in DI versus 2.4% in SI) and about 15 pp is common-cut failure from shifted kinematics. The isolation stage is barely affected (−3 pp), since the extra cone energy is small compared with the 4 GeV threshold. The photon-ID stage loses 13 pp, entirely driven by the vertex-displaced shower-shape inputs feeding the E_T -dependent BDT threshold: the vertex-shifted population has a BDT pass rate of only $\sim 60\%$ versus 82.7% for the unshifted population. The integrated DI-induced efficiency reduction is about 9% at 0 mrad and about 3% at 1.5 mrad, scaling with the cluster-weighted pileup fraction.

These effects are absorbed directly into the nominal pipeline rather than applied as a separate correction: the efficiency, the response matrix, and the inclusive-jet background template are all built from the SI+DI mixed ensemble, with each event carrying the weight $w = \sigma_{\text{eff}} \cdot f_{\text{mix}} \cdot \text{TVR}(z_h, z_{mb})$, where $f_{\text{mix}} = 1 - f_{\text{DI}}$ for SI samples and f_{DI} for DI samples, and TVR is the truth-vertex reweight of Appendix K. The single free parameter f_{DI} is varied to obtain the systematic of item 5.8.

Shower-shape data–MC validation. Before fitting the blend fraction, the model is validated at the sample level. The most pileup-sensitive observable is the center-of-gravity width $w_{\text{CoG}\eta}$, which is broadened directly by the vertex shift via the $\Delta\eta$ relation above; the photon-ID BDT score amplifies the residual through its nonlinear discriminant, while the energy-spread-driven observables (f_1 , $E_{1\times 1}/E_{3\times 3}$) are far less DI-sensitive. The original Figs. 75–76 overlay pure SI MC, pure DI

Table 11: RMS of Δz for single and double interactions [cm].

Class	RMS Δz [cm]
Single	7.6
Double	34.5

Table 12: Per-stage efficiency [%] for single, double and mixed events, and the net effect (percentage points).

Stage	Single	Double	Mixed	Effect
$\varepsilon_{\text{reco}}$	93.5	73.1	88.4	−20 pp
ε_{iso}	74.2	71.0	73.5	−3 pp
ε_{id}	82.7	69.4	80.0	−13 pp
ε_{all}	57.4	36.0	52.0	−21 pp

MC, and data from both crossing-angle periods, all area-normalized at the preselection cut. The 0 mrad data systematically track the DI tail, while the 1.5 mrad data track closer to the SI MC — exactly as expected from their 22.4% versus 7.9% pileup fractions — providing a direct sample-level confirmation of the blending picture. Representative data-inclusive-MC χ^2/ndf values for the two leading discriminants at the lowest and highest E_T^γ bins are given in Table 13; the mid- and high- E_T^γ bins improve dramatically once the DI component is blended in, whereas the lowest bin is dominated by a residual DI-MC mismodeling discussed below.

Data-driven blend-fraction scan. The nominal blending uses the analytic cluster-weighted fractions $f_{\text{DI}} = 0.224$ (0 mrad) and 0.079 (1.5 mrad). The systematic uncertainty on f_{DI} is sized by treating the fraction as a free parameter and minimizing the χ^2 between the data and an inclusive MC rebuilt by shape interpolation,

$$\text{MC}_{\text{incl}}(f) = (1 - f) T_{\text{SI}} + f T_{\text{DI}}. \quad (23)$$

The pure SI and DI templates $T_{\text{SI}}, T_{\text{DI}}$ are recovered by “de-baking” the analytic weights out of the combined hadds, $T_{\text{SI}} = \text{SI}_{\text{total}}/(1 - f_{\text{DI}}^{\text{anal}})$ and $T_{\text{DI}} = \text{DI}_{\text{total}}/f_{\text{DI}}^{\text{anal}}$, and the round-trip identity $(1 - f)T_{\text{SI}} + fT_{\text{DI}} = \text{SI}_{\text{total}} + \text{DI}_{\text{total}}$ at $f = f_{\text{DI}}^{\text{anal}}$ is verified numerically. The scan is a one-parameter shape-only χ^2 against the data BDT-score distribution at the preselection cut (NPB-clean, jet-dominated), unit-area-normalized in each E_T^γ bin so that the absolute jet cross-section uncertainty does not leak into f_{DI} . The BDT score is chosen because it aggregates the eleven shower-shape inputs through a discriminant already trained on the SI/DI separation. Each crossing angle is scanned over $f \in [0.00, 0.50]$ in steps of 0.005, summing residuals over five E_T^γ cells ($5 \times 50 = 250$ bins). The before/after χ^2/ndf for the two variables and two E_T^γ ranges is given in Table 13, and the scan minimum in Table 14.

The 0 mrad scan has a clean parabolic minimum at $f_{\text{best}} = 0.290$, above the analytic anchor 0.224; the 1.5 mrad minimum sits at the lower scan edge $f_{\text{best}} = 0$, below the analytic anchor 0.079. The residual χ^2/ndf at the minimum is far from unity (2.57 and 3.20), so the $\Delta\chi^2 = 1$ prescription does not give a statistically meaningful interval; moreover the two crossings pull f_{best} in *opposite* directions relative to their anchors, whereas a coherent DI-shape mismodeling would shift them in the same fractional direction. This indicates that the χ^2 surface is dominated by data-MC tensions unrelated to the DI fraction (truth-vertex-reweight closure, jet-fragmentation

modeling, BDT calibration). Accordingly the central prediction is kept at the analytic anchors, and the fit is used only to size the envelope: a single variation per crossing angle is run at f_{best} (0.290 at 0 mrad, 0.000 at 1.5 mrad), and the absolute per-bin cross-section deviation is taken as the symmetric two-sided systematic of item 5.8.

Table 13: χ^2/ndf before $\rightarrow$ after DI blending, for two variables and two E_T^γ ranges, per crossing-angle configuration.

Variable	E_T^γ range	0 mrad	1.5 mrad
$w_{\text{CoG}\eta}$	10–14	47.40 $\rightarrow$ 28.96	9.72 $\rightarrow$ 42.31
$w_{\text{CoG}\eta}$	22–28	15.58 $\rightarrow$ 0.75	21.10 $\rightarrow$ 1.31
BDT	10–14	6.85 $\rightarrow$ 20.30	5.98 $\rightarrow$ 33.08
BDT	22–28	9.62 $\rightarrow$ 1.75	1.89 $\rightarrow$ 1.13

Table 14: DI blend-fraction scan: analysis value, best-fit value, before/after χ^2/ndf and $\Delta\chi^2$.

Quantity	0 mrad	1.5 mrad
f_{anal}	0.224	0.079
f_{best}	0.290	0.000
χ^2/ndf (before)	2.57	3.20
χ^2/ndf (after)	2.84	3.67
$\Delta\chi^2$	68.4	116.6

Cluster size under DI. The EMCal cluster size (the owned-tower count n_{owned} and the bounding-box widths $\Delta i_\eta, \Delta i_\phi$) is the basic shape property underlying every shower-shape discriminant. The DI-induced shifts are mild for photons ($\Delta\langle n_{\text{owned}} \rangle \sim 0.2$ towers across the analysis range) but several times larger for jets, because soft neighboring towers from the second collision can be pushed above the 70 MeV clustering threshold in a busy jet environment. After the tight photon-ID selection, data, jet MC, and photon MC all converge to a signal-like cluster size within $\pm 5\%$, so the residual DI effect on the ABCD background template is most visible at the preselection stage and is reduced to below about 1% after the tight cut.

Energy response under DI. The photon energy response $R = E_T^{\text{reco}}/p_T^{\text{truth}}$ is measured in bins of truth p_T across the SI, DI, and mixed samples by fitting a double-sided Crystal-Ball function to extract the peak μ and resolution σ . At the common-cut level the DI events shift the peak by $\lesssim 0.5\%$ and broaden the resolution by $\lesssim 10\%$ at $p_T \sim 20$ GeV, with the mixed curves sitting between SI and DI at the appropriate fractions. This shift is well inside the energy-scale systematic budget and is further reduced by the truth-vertex reweight, so the leading DI sensitivity remains on the efficiency and the background template rather than on the energy scale.

DI reco–truth kernel. A structural feature limits any truth-only vertex correction for DI events. In `jet12_double` MC at 1.5 mrad the conditional reco kernel $P(z_r | z_h = 0)$, with the minimum-bias vertex z_{mb} marginalized, has an RMS of ~ 34 cm — wider than the ~ 22 cm data target. Because the reconstructed vertex of a DI event is the average of two independently drawn vertices, even forcing the primary truth vertex to $z_h = 0$ cannot narrow the reco distribution below this

kernel width. This is the geometric reason a reweight on the primary truth vertex z_h alone is insufficient for DI events, and it motivates the two-vertex $w(z_h)w(z_{mb})$ factorization adopted in the next appendix.

K Truth-vertex reweighting

The longitudinal extent of the luminous region differs between the two crossing-angle periods ($\sigma_{\text{lum}} \approx 54$ cm at 0 mrad, ≈ 22 cm at 1.5 mrad) and from the common generator beam profile ($\sigma_{\text{gen}} \approx 65$ cm). Since the vertex-cut acceptance and the vertex-driven shower shapes both depend on the z distribution, the MC truth-vertex distribution must be reweighted to match data. A naive reco-level weight $f(z_r)$ is insufficient for DI events: $f(z_r)$ depends only on the symmetric coordinate $\bar{z} = (z_h + z_{mb})/2$ and is flat in the antisymmetric coordinate $\varepsilon = (z_h - z_{mb})/2$, so the antisymmetric component is unconstrained by reco closure alone and the primary truth vertex retains a width $\sigma(z_h) = \sqrt{\frac{1}{2}\sigma_{\text{lum}}^2 + \frac{1}{2}\sigma_{\text{MC}}^2} \simeq 47$ cm at 1.5 mrad. Left uncorrected, this biases the DI truth fiducial acceptance low by about 19 pp, propagating to a +1.4% bias on the cross section at 1.5 mrad (and $\sim 0.3\%$ at 0 mrad). The reweight is therefore factorized as $w(z_h)$ for SI events and $w(z_h)w(z_{mb})$ for DI events, giving the iteration access to the antisymmetric coordinate.

Method. The weight function $w(z)$ is derived directly from the data reco distribution with no closed-form Gaussian assumption. Under a candidate weight, the mixed-MC reco distribution is built event-by-event,

$$M(z_r) = f_s \sum_{e \in T_s} w(z_h^e) \delta(z_r - z_r^e) + f_d \sum_{e \in T_d} w(z_h^e) w(z_{mb}^e) \delta(z_r - z_r^e), \quad (24)$$

where f_d is the cluster-weighted DI fraction and $f_s = 1 - f_d$; for DI events the weight enters once per truth vertex. Both periods use 80 bins over $[-200, 200]$ cm, with the closure window auto-derived as the 99th percentile of $|z_r^{\text{data}}|$ ($z_{\text{cut}} = 83$ cm at 1.5 mrad, 144 cm at 0 mrad). The seed is $w_0(z) = \mathcal{N}(0, \sigma_{\text{seed}}^2) / \mathcal{N}(0, \sigma_{\text{gen}}^2)$ with $\sigma_{\text{gen}} = 64.997 \pm 0.015$ cm from the MC truth vertex and $\sigma_{\text{seed}} = \text{std}(z_r^{\text{data}})$ (21.48 / 54.32 cm); σ_{seed} needs no physics input — it is simply the empirical width of the distribution being matched. The update is Lucy–Richardson-like, multiplicative, and damped,

$$z_{\text{reco}}^{\text{model}} = \mathcal{F}(z_{\text{truth}}), \quad (25)$$

$$w_{i+1}(z) = w_i(z) \cdot [\text{smooth}(r_i(z))]^\alpha, \quad \alpha = 0.5, \quad (26)$$

with $r_i(z_r) = D(z_r)/M_i(z_r)$ the data/MC residual applied at the matching truth bin. After each step w is renormalized so $\langle w(z_h) \rangle = 1$ over the SI truth distribution (preserving the physics DI fraction); one-bin Gaussian smoothing suppresses oscillation, the update factor is clipped to $[0.5, 2]$ per iteration, and boundary bins are edge-clamped. Convergence is declared at $\max |r_i(z) - 1| < 0.01$ inside the window, with a 30-iteration cap. The converged parameters are listed in Table 15.

Per-period results. Both periods are fit on the `photon10/photon10_double` mix; the headline closure numbers are in Table 15. The two cases behave very differently. At 0 mrad the luminous region ($\sigma_{\text{lum}} \approx 54$ cm) is already close to the generator profile and the data reco shape is nearly Gaussian, so the seed alone gives $\max |r - 1| \approx 8\%$ and the iteration converges cleanly in 8 steps to $\max |r - 1| \approx 0.8\%$ and $\chi^2/\text{ndf} \approx 2.78$; the recovered $w(z)$ is smooth and broadly Gaussian with a small central enhancement. At 1.5 mrad the narrow luminous region must be extracted from

Table 15: Truth-vertex-reweighting parameters by crossing-angle configuration.

Quantity	0 mrad	1.5 mrad
Luminosity [pb ⁻¹]	47.2076	17.1642
f_{double}	0.224	0.079
σ_{seed} [cm]	54.32	21.48
σ_{gen} [cm]	64.997 ± 0.015	
Final χ^2/ndf	2.78 (ndf 57)	13.7 (ndf 33)
max $ r - 1 $	0.008	0.111

a roughly three-times-wider source, so $w(0) \approx 8$, and the data reco distribution is non-Gaussian (a narrow core plus a mid-tail shoulder near $|z_r| \approx 45\text{--}60$ cm). The iteration makes dramatic progress in the first ~ 5 steps (the in-window χ^2 falls by a factor ~ 3100) and then enters a slow plateau driven by bins at the closure-window edge $|z_r| \approx 83$ cm, where the wide DI reco-truth kernel (previous appendix) makes the residual insensitive to $w(z)$. After 30 iterations the bulk $|z_r| < 30$ cm is closed to 1–2% per bin while the edge residual stalls at $\max |r - 1| \approx 0.11$. Inside the analysis fiducial $|z| < 60$ cm the per-bin closure is at the 1–2% level for both periods.

A single-Gaussian ansatz cannot capture the 1.5 mrad shape: a parametric σ_{tgt} scan bottoms out at $\sigma_{\text{tgt}} \approx 15.7$ cm with a χ^2/ndf two orders of magnitude worse than the iterative result, whereas at 0 mrad the parametric and iterative solutions nearly coincide because the data is already Gaussian. The recovered 1.5 mrad $w(z)$ has three distinguishing features that no symmetric Gaussian can represent: a strong central peak ($w(0) \approx 8$) pulling near-IP doubles into the data-like core, a mid- z dip suppressing the generator’s natural $|z| \sim 30\text{--}50$ cm population, and a clear $\pm$ asymmetry in $|z| \approx 40\text{--}60$ cm tracking the asymmetric data mid-tail. The reweight is fit on the photon mix; extending it to the full sample blend (including `jet12`) is expected to shift the closure metrics by about 1%. The residual cross-section impact, estimated from the maximum-deviation iteration, is $\mathcal{O}(1\text{--}2\%)$, concentrated at low E_T^γ where the vertex-cut acceptance is most sensitive; it is carried as a cross-check and absorbed into the photon-ID response-driven systematic group rather than added in quadrature.

L herwig cross-check

The efficiency factors of Sec. 4 are derived from PYTHIA-8 MC and therefore inherit any mismodeling of the event topology around the photon — in particular the underlying-event and beam-remnant activity that sets the isolation and MBD-vertex efficiencies. To bound this generator dependence, the full 1.5 mrad analysis pipeline is rerun with the nominal PYTHIA-8 Detroit signal MC replaced by HERWIG 7 photon-jet samples (`photon10_herwig`, `photon20_herwig`), keeping the data input and all background MC unchanged. HERWIG uses an angular-ordered parton shower and a cluster hadronization model, in contrast to PYTHIA’s dipole shower and Lund string fragmentation, so the comparison probes a genuinely different modeling of the soft environment. The truth- p_T prior is rederived in the same way as the nominal (an unweighted single-iteration unfolding of the data with the HERWIG response matrix, refit to a rational form), and the unfolded comparison uses a common reweight target so that any residual difference reflects the response matrix alone rather than the prior choice.

The differences are concentrated in the isolation and MBD-vertex efficiencies: HERWIG photons sit in less-active mid-rapidity environments at fixed truth E_T^γ , raising the isolation efficiency by

3–6% and the MBD-vertex efficiency by 4–7%. The photon-ID efficiency, which is governed by the calorimeter shower physics rather than the event topology, agrees to within $\pm 1.5\%$ across the analysis range, and the unfolded yield (before efficiency division) collapses to within $\pm 4\%$ once the common reweight target is used. The isolation-efficiency difference is propagated as the generator-driven systematic in the efficiency group (Sec. 5, item 5.4.2, $\pm 5.3\%$). The MBD-vertex difference is *not* double-counted here, because the data-driven MBD-coincidence efficiency from the PPG-10 luminosity note already covers the generator-modeling uncertainty on the MBD-trigger response (item 5.5). The unfolded-yield comparison is retained as a closure cross-check only.